

WeGlow — Phase 3 (Navigation) Audit

Branch `phase-3-navigation` · Reviewed 2026-09-09 · Build: `clean architectureCheck testDebugUnitTest lintDebug assembleDebug` — BUILD SUCCESSFUL · Manual navigation: no crashes reported

Final verdict

PHASE 3 FAIL

Two real blockers must be fixed before Phase 4. Both are **incorrect back-stack behavior**, not compile or runtime failures — which is consistent with the build passing and no crashes during manual testing.

Scope reviewed: `WeGlowApp.kt` (working-tree version), `Destination.kt`, `AuthViewModel`, `OnboardingViewModel`, `ScanViewModel`, and `WeGlowBottomNavigation.kt` — against navigation architecture, startup routing, login/signup back stack, onboarding flow, bottom navigation, scan/results navigation, logout navigation, `launchSingleTop`, state restoration, and back-stack handling.

Blockers

Blocker 1 — Onboarding screens stay on the back stack after entering Home

`WeGlowApp.kt:414-425` — `WelcomeIntro` → Home

```
navController.navigate(Destination.Home.route) {
    popUpTo(Destination.WelcomeIntro.route) { inclusive = true }
    launchSingleTop = true
}
```

This pops only `welcome_intro`. Every entry to Home from onboarding — post-signup, sign-in of a not-yet-onboarded user, and startup `ONBOARDING` routing — lands with the stack:

```
[age_selection, skin_type, gender_selection, skin_sensitivity, home]
```

System back from Home re-enters the onboarding wizard at `skin_sensitivity`. This is a regression: the diff changed the correct `popUpTo(Destination.AgeSelection.route) { inclusive = true }` (which cleared the whole wizard) to `popUpTo(WelcomeIntro)`.

Fix: restore `popUpTo(Destination.AgeSelection.route) { inclusive = true }` (or pop the graph start inclusive) so Home is the base of the stack after onboarding.

Blocker 2 — Bottom navigation pops up to a destination that isn't on the back stack

WeGlowApp.kt:99–108 (bottom nav) and WeGlowApp.kt:541–555 (ScanResults → Routines)

```
popUpTo(navController.graph.findStartDestination().id) { saveState = true }
launchSingleTop = true
restoreState = true
```

`findStartDestination()` resolves to `loading`, which is removed from the back stack at startup (`popUpTo(Destination.Loading.route) { inclusive = true }`, lines 166-170). In Navigation-Compose 2.8.0, `popUpTo` to an id not present on the back stack pops nothing. Consequences:

- Switching tabs **accumulates** entries: Home → Discover → Scan → Routines → Profile produces `[home, discover, scan, routines, profile]`. `launchSingleTop` only de-dupes the current top, so even revisiting a tab stacks a duplicate.
- Back button from any tab walks backward through every previously visited tab instead of returning to Home / exiting.
- `saveState` / `restoreState` never engage as intended because the pop they depend on never happens — tab state restoration is effectively dead.

Fix: give the main area a stable anchor and pop to it, e.g. `popUpTo(Destination.Home.route) { saveState = true }` (works once Blocker 1 makes `home` the base), or introduce a nested graph for the authenticated area with its own start destination and pop to that.

Passed — no action needed

- Startup routing (`Loading` → login/onboarding/home, `loading` popped inclusive, `launchSingleTop` guard) — correct.
- Login → Signup back stack, and `popUpTo(Login) { inclusive = true }` on sign-in / sign-up success — correct.
- Scan → ScanResults / HairstyleResults with `popUpTo(Scan) { inclusive = true }`; onBack via `popBackStack()` — correct.
- Logout (`popUpTo(navController.graph.id) { inclusive = true }` + `scanViewModel.clear()`) — correctly clears the entire stack to `[login]`.
- `launchSingleTop` is applied consistently on forward navigations.

Minor — not blocking, worth tidying in Phase 3

- `HomeScreen` `onScanClick` / `onDiscoverClick` push `scan` / `discover` as plain navigations (no `popUpTo` / `restoreState`), so reaching a tab via a Home shortcut vs. the bottom bar creates divergent stacks and duplicate tab entries.
- `age_selection` has no back affordance; system back from it exits the app.